

User Manual: Grocery Sales Forecasting System

1. Introduction

The Grocery Sales Forecasting System is a robust machine learning pipeline designed to predict future store sales using LightGBM. It integrates relational store metadata, economic indicators (oil prices), and holiday schedules to deliver time-aware, accurate sales forecasts.

2. System Requirements

- Python 3.8 or higher
- At least 16GB RAM (for processing large time-series data)
- Standard data science stack (pandas, numpy, lightgbm)
- Storage: Sufficient space for raw CSV data and generated feature matrices

3. Setup & Installation

3.1 Repository Setup

1. Clone the project repository to your local machine.
2. Ensure your data directory exists (named 'data') and contains the raw CSV files: train.csv, test.csv, stores.csv, oil.csv, holidays_events.csv.

3.2 Installation

Install the required Python packages:

```
pip install pandas numpy lightgbm
```

4. Executing the System

The forecasting system consists of three main stages executed via the pipeline.

4.1 Training & Inference

To run the full end-to-end training and inference process, execute the compute.py script:

```
python compute.py
```

This script performs the following tasks:

- Ingests and merges relational data from the 'data' directory.
- Generates complex temporal, lag, rolling, and promotion-based features.

- Performs a time-aware validation split.
- Trains a LightGBM model and calculates RMSLE.
- Exports feature importance metrics and generates a submission.csv file.

5. System Components

`data_pipeline.py`: Handles data ingestion, table merging, and oil price imputation.

`features.py`: Contains the `FeatureEngineer` class, responsible for generating lags, rolling averages, and categorical encodings.

`validation.py`: Implements `TimeAwareValidation` to ensure testing mimics real-world scenarios.

`compute.py`: Orchestrates the training, validation, feature importance analysis, and final inference.